SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 10%

QUESTIONS: 5 Total Marks:50

Annexure A

TECHNICAL PRESENTATION

Code of Conduct:

I P.Nikitha/192324226(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

P. Nikitha

Signature of Student:

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Saveetha Institute of Medical And Technical Sciences

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TECHNICAL PRESENTATION

SUBSTITUTION vs TRANSPOSITION

Which Provides Better Security?

CSA5112-CRYPTOGRAPHY AND NETWORK SECURITY

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1. What Is Cryptography?

The science of protecting information from unauthorized access.

PLAINTEXT

The original readable message.
Example: ATTACK AT DAWN

→

ENCRYPTION

A method and key transform the message.

→

CIPHERTEXT

The scrambled output.
Example: a coded message

- Encryption protects confidentiality; other cryptographic tools can support integrity and authentication.
- Classical ciphers help us learn the basics, but are not secure for modern sensitive data.

2. Substitution Cipher

Replace each plaintext symbol with another symbol.

HOW IT WORKS

The letter changes, but its position in the message stays the same.

CAESAR CIPHER (+3)

A → D, B → E, C → F …
Each letter is shifted three places forward.

EXAMPLE :

Plaintext: H E L L O

Ciphertext: K H O O R

The letters are replaced; their order is unchanged.

3. Transposition Cipher

Rearrange the positions of symbols without changing the symbols themselves.

HOW IT WORKS

The letters stay the same, but their order changes according to a rule or key.

SIMPLE REVERSAL EXAMPLE

Plaintext: HELLO
Reverse the order → OLLEH

LETTER POSITIONS



The same letters are present, but their positions have changed.

4. Side-by-Side Comparison

The key difference is what each method changes.

Feature	Substitution	Transposition
Main operation	Replaces symbols	Rearranges positions
Letters	Usually change	Stay the same
Order	Usually preserved	Changes
Classic example	Caesar cipher	Rail Fence cipher
Common weakness	Letter-frequency patterns	Patterns in position / structure

Both are classical techniques and can be combined in a product cipher.

5. What Does “Better Security” Mean?

Security is more than making a message look scrambled.

KEY SPACE

How many possible keys exist? More possibilities can make guessing harder.

PATTERN LEAKAGE

Does the ciphertext reveal letter frequencies or message structure?

IMPLEMENTATION

Is the method correctly designed and used with a strong key?

- A simple Caesar cipher has only 25 non-trivial shifts, so it is easy to try every key.
- A simple rearrangement cipher also preserves clues and can often be solved with analysis.
- Neither classical method alone is suitable for protecting modern confidential data.

6. Which Is Easier to Break?

A small worked example shows why both simple methods are weak.

SUBSTITUTION: CAESAR +3

Plaintext: HELLO
Ciphertext: KHOOR
An attacker can test the 25 possible shifts and look for readable text.

TRANSPOSITION: REVERSAL

Plaintext: HELLO
Ciphertext: OLLEH
If the rule is simply “reverse,” the attacker reverses it back.

LESSON

A cipher can look confusing but still be easy to break when its method is simple and predictable.

7. Can We Combine Them?

Classical cryptography often combines transformations to make analysis harder.

STEP 1

Start with plaintext:
HELLO

→

STEP 2

Substitute letters:
Caesar +3 → KHOOR

→

STEP 3

Transpose positions:
Reverse → ROOHK

- This illustrates combining substitution and transposition; it is still only a classroom example.
- Modern secure systems use carefully analyzed algorithms, not these simple hand ciphers.

Step-by-Step Example 1: Ceasar Cipher

Caesar cipher with a shift key of +3

Goal: Encrypt the word MANGO

STEP 1 — KEY

Shift each letter 3 places forward in the alphabet.

STEP 2 — M

M →N → O → P, so M becomes P.

STEP 3 — A

A → B → C → D, so A becomes D.

STEP 4 — N

N → O → P → Q, so N becomes Q.

STEP 5 — G

Repeat the same rule: G becomes J.

STEP 6 — O

O → P → Q → R, so O becomes R.

Plaintext: H E L L O → Ciphertext: P D Q J R

Remember: the letters change, but their positions stay the same.

Monoalphabetic Cipher: Fixed Letter Mapping

Every plaintext letter always maps to the same ciphertext letter throughout the message.

Plain alphabet: ABCDEFGHIJKLMNOPQRSTUVWXYZ
Cipher alphabet: QWERTYUIOPASDFGHJKLZXCVBNM

1 Find B

B maps to W in the cipher alphabet.

2 Find R

R maps to K.

3 Find A

A maps to Q.

4 Finish BRAVE

V→C and E→T.
Join: W K Q C T.

PLAINTEXT		CIPHERTEXT
BRAVE	→	**WKQCT**

Unlike Caesar, the mapping need not be a simple shift. A fixed mapping can still leak letter-frequency patterns.

Vigenère Cipher: Repeating Keyword

Each letter is shifted by a different amount determined by a repeating keyword.

Plaintext: ATTACK • Key: KEYKEY • Use A=0, B=1, …, Z=25.

1 Align key

Plaintext: A T T A C K
Key: K E Y K E Y

2 First pair

A + K(10) = K

3 Next pairs

T+E(4)=X
T+Y(24)=R

4 Finish

A+K=K
C+E=G
K+Y=I

PLAINTEXT		CIPHERTEXT
ATTACK	→	**KXRKGI**

The keyword changes the shift from letter to letter. This is a classical example, not modern secure encryption.

Hill Cipher: Matrix-Based Encryption

Letters are converted to numbers and encrypted in blocks using matrix multiplication modulo 26.

Key K = [[3, 3], [2, 5]] • A=0, B=1, …, Z=25 • Encrypt HELP in pairs.

1 Pair 1: HE

H=7, E=4
$K \times [7,4]^T = [33,34]^T$
mod 26 → [7,8] = HI

2 Pair 2: LP

L=11, P=15
$K \times [11,15]^T = [78,97]^T$
mod 26 → [0,19] = AT

3 Combine

First block: HI
Second block: AT

4 Check

33 mod 26=7 (H)
34 mod 26=8 (I)
78 mod 26=0 (A)
97 mod 26=19 (T)

PLAINTEXT		CIPHERTEXT
HELP	→	**HIAT**

The Hill cipher encrypts letter blocks together. The key matrix must be invertible modulo 26 for decryption to work.

Affine Cipher: Multiply and Add

A mathematical substitution cipher using a multiplication value and an additive value.

Encryption: E(x)=(5x+8) mod 26 • A=0, B=1, …, Z=25.

1 Encrypt W

W=22
(5×22+8) mod 26
=118 mod 26 =14 → O

2 Encrypt O

O=14
(5×14+8) mod 26
=78 mod 26 =0 → A

3 Encrypt R

R=17
(5×17+8) mod 26
=93 mod 26 =15 → P

4 Finish WORLD

L→L and D→X.
Ciphertext: O A P L X

PLAINTEXT		CIPHERTEXT
WORLD	→	OAPLX

The multiplier 5 is coprime with 26, so it has a modular inverse and the mapping can be reversed with the key.

Playfair Cipher: Encrypt Letter Pairs

Playfair uses a 5×5 key square and encrypts pairs of letters rather than single letters.

Keyword: MONARCHY • I/J share one square • Plaintext: INSTRUMENTS (pad final S with X).

1 Key square

M O N A R
C H Y B D
E F G I/J K
L P Q S T
U V W X Z

2 Make pairs

IN | ST | RU
ME | NT | SX

3 Apply rules

Same row: move right.
Same column: move down.
Rectangle: use other corners in each row.

4 Encrypt pairs

IN→GA, ST→TL,
RU→MZ, ME→CL,
NT→RQ, SX→XA

PLAINTEXT		CIPHERTEXT
INSTRUMENTS	→	GATLMZCLRQXA

Playfair substitutes digraphs (letter pairs). X is padding here; the exact output depends on the keyword square and pairing rules.

Step-by-Step Example 2: Transposition

Simple reversal method — an introductory transposition example

Goal: Encrypt the word HELLO

STEP 1 — WRITE

Write the plaintext: H E L L O

STEP 2 — RULE

Reverse the order of all letters.

STEP 3 — REORDER

Read from right to left: O L L E H

Before

H E L L O → O L L E H

After

Plaintext: HELLO → Ciphertext: OLLEH

Remember: the letters remain the same, but their positions change.

Rail Fence Cipher: 2 Rails

Write the plaintext by alternating letters between two rows, then read across each row.

Plaintext: WEAREDISCOVERED • Key: 2 rails • Ignore spaces while encrypting.

1 Write odd positions

Top rail:
W A E I C V R D

2 Write even positions

Bottom rail:
E R D S O E E

3 Read top row

WAEICVRD

4 Read bottom + join

ERDSOEE
Top + bottom gives ciphertext.

PLAINTEXT
WEAREDISCOVERED → **CIPHERTEXT**
WAEICVRDERDSOEE

All letters are preserved; only their positions change. Rail Fence is a transposition cipher.

Columnar Transposition: Read Columns by Key

Write the message in rows under a numbered key, then read columns in ascending key order.

Plaintext: ATTACKATDAWN • Key labels left-to-right: 3 1 2.

1 Fill the grid

Key: 3 1 2
A T T
A C K
A T D
A W N

2 Read key 1

Key 1 is column 2:
T C T W → TCTW

3 Read key 2

Key 2 is column 3:
T K D N → TKDN

4 Read key 3

Key 3 is column 1:
A A A A → AAAA

PLAINTEXT		CIPHERTEXT
ATTACKATDAWN	→	TCTWTKDNAAAA

Ciphertext is column 2 + column 3 + column 1. No padding is needed because 12 letters fill the 3-column grid evenly.

9. Conclusion

So, which provides better security?

SUBSTITUTION

Changes the symbols.
Example: HELLO → KHOOR (Caesar +3)
Weak if the replacement rule is simple.

TRANSPOSITION

Changes the order.
Example: HELLO → OLLEH (reversal)
Weak if the rearrangement rule is simple.

FINAL CONCLUSION

Neither is inherently more secure. Security depends on the algorithm, key strength, and resistance to cryptanalysis. Modern systems use rigorously tested cryptographic algorithms.

THANK YOU

RUBRICS FOR CALCULATION:

Criteria	Weigh ta ge	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understa ndi ng of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/techno lo gies	Good understa ndi ng with proper integration of most concepts	Basic understan din g; limited or partial integration	Poor understan din g; unable to integrate concepts
Experim ent al Design	30	Well-planned, systematic implementati on with optimal solution	Correct implemen tat ion with minor issues	Basic implementa ti on with errors or inefficiencies	Incorrect or incomplete implement ati on
Use of Tools	20	Effective and advanced use of tools/technologi es with proper configuration	Appropria te use of tools with correct execution	Limited or inconsiste nt use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonab le interpreta tio n	Basic analysis with limited insights	Poor or incorrect analysis of results
Documen tat ion	10	Clear, well structured documentation with diagrams, code, and results	Organized document ati on with necessary details	Basic documenta tio n with missing details	Poor or incomplete documenta tio n